# Swastik Majumder

Curriculum Vitae

Department of Physics, The Ohio State University

## Education

**Doctor of Philosophy in Physics** 2025–Present
The Ohio State University, Columbus, Ohio, United States.

**Bachelor of Science and Master of Science in Physics** 2020–2025
Indian Institute of Science Education and Research Kolkata, India.

## Research Interests

**In-medium quantum field theory:** Perturbative quantum chromodynamics at finite temperature and density through effective field theories.

**Statistical mechanics:** Equilibrium and nonequilibrium behavior of classical many body systems.

## Publications

5. **S. Majumder** and T. Gorda.
Hot and dense QCD at $N^3LO$: infrared sector.
*In preparation* (2026).

4. **S. Majumder** and T. Gorda.
Hot and dense QED at $N^3LO$: infrared sector.
*Preprint* (2026).
arXiv:2610.10771.

3. **S. Majumder** and M. Barma.
Stochastic systems with Bose-Hubbard interactions: Effects of bias on particles on a random comb.
*Physical Review E* **113**, 044137 (2026).
Journal | arXiv:2601.09962.

2. **S. Majumder** and M. Barma.
Stochastic systems with Bose-Hubbard interactions: Effects of bias on particles on a one-dimensional lattice.
*Physical Review E* **113**, 044136 (2026).
Journal | arXiv:2508.03344.

1. **S. Majumder** and N. Sasakura.
Three cases of complex eigenvalue/vector distributions of symmetric order-three random tensors.
*Progress of Theoretical and Experimental Physics* **2024**, 093A01 (2024).
Journal | arXiv:2408.01030.

## Presentations

**Poster presentation** October 2026
**S. Majumder** and T. Gorda.
*Thermal corrections to the equation of state of dense QED at high perturbative orders.*
American Physical Society Division of Nuclear Physics Fall Meeting, Philadelphia, Pennsylvania.

**Poster presentation** August 2026
**S. Majumder** and T. Gorda.
*The $N^3LO$ Mixed Contribution to the Pressure of a Hot and Dense Relativistic QED Plasma.*
Department of Physics Poster Session, The Ohio State University, Columbus, Ohio.

## Research Experience

**The Ohio State University** 2026–Present
*Graduate Research Associate*, research supervisor: Dr. Tyler Gorda.

**Tata Institute of Fundamental Research, Hyderabad** 2024–2025
*Visiting Research Student*, research supervisor: Prof. Mustansir Barma.

**Yukawa Institute for Theoretical Physics, Kyoto University** 2023–2024
*Visiting Research Student*, research supervisor: Dr. Naoki Sasakura.

## Teaching Experience

**The Ohio State University** — *Graduate Teaching Associate*

Electricity and Magnetism, Waves, Optics, and Modern Physics Spring 2026, Autumn 2025

**Indian Institute of Science Education and Research Kolkata** — *Teaching Assistant*

Statistical Mechanics Spring 2025, Spring 2024

Probability and Statistics II Spring 2025

Condensed Matter Physics Autumn 2024

## Academic Service

**President, Physics Graduate Student Council** 2025–2026
The Ohio State University.

**Leadership team, Science Club** 2021–2022
Indian Institute of Science Education and Research Kolkata.

## Awards and Honors

**Graduate Associate Teaching Award nomination** 2025
The Ohio State University.

**Silver Medal,** University Physics Competition 2022

**Finalist,** International Astronomy and Astrophysics Competition 2021

## References

**Dr. Tyler Gorda**
Department of Physics and Center for Cosmology and AstroParticle Physics (CCAPP),
The Ohio State University, United States.

**Prof. Mustansir Barma**
Tata Institute of Fundamental Research, Hyderabad, India.

**Dr. Naoki Sasakura**
Yukawa Institute for Theoretical Physics, Kyoto University, Japan.

**Prof. Arindam Kundagrami**
Department of Physical Sciences,
Indian Institute of Science Education and Research Kolkata, India.

## Software and Programming

Python, Mathematica, LaTeX.

## Languages

**Bengali:** mother tongue.

**Hindi and English:** bilingual proficiency.